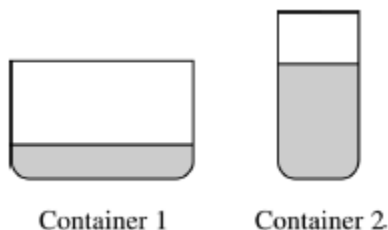


Chapter 10 Part 2: Le Chatelier's Principle MCQ

1.



The figure above shows two closed containers. Each contains the same volume of acetone in equilibrium with its vapor at the same temperature. The vapor pressure of the acetone is

- (A) higher in container 1 because the surface area of the liquid is greater
- (B) higher in container 1 because the volume of vapor is greater
- (C) lower in container 1 because the level of the liquid is lower
- (D) the same in both containers because the volume of the liquid is the same

(E) the same in both containers because the temperature is the same



2. Which of the following statements is true for the equilibrium vapor pressure of a liquid in a closed system?

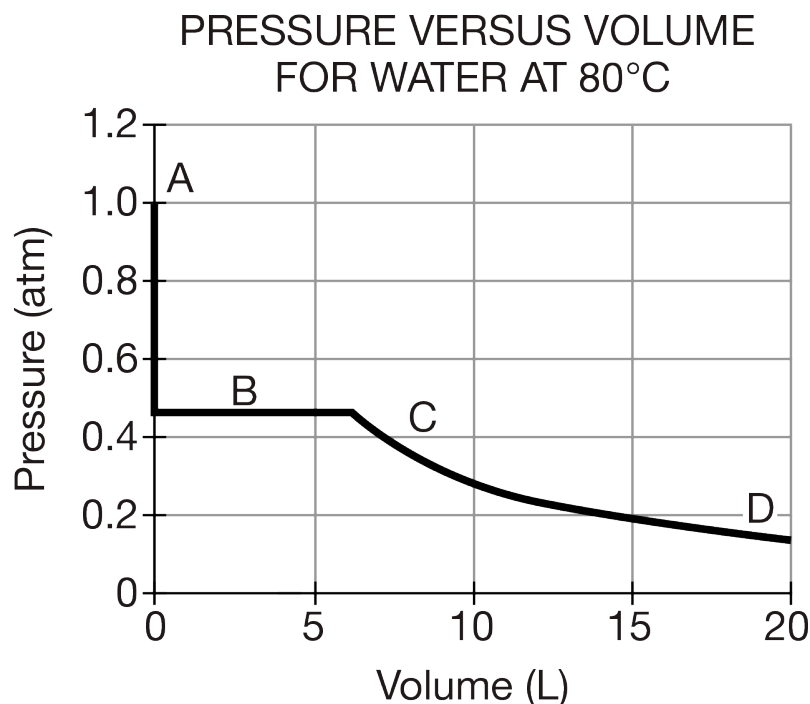
- (A) It remains constant when the temperature increases.
- (B) It decreases to half its original value if the volume of the gas phase is doubled.
- (C) It increases to twice its original value if the volume of the liquid phase is doubled.
- (D) It decreases to half its original value if the surface area of the liquid is reduced by one-half.

(E) It is independent of the volume of the vapor phase.

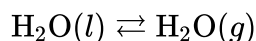


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3.



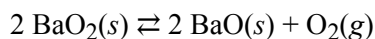
A cylinder with a moveable piston is completely filled with a small amount (100 millimoles) of liquid water at a pressure of 1.0 atm and a temperature of 80°C. All the air in the cylinder is excluded. The cylinder is placed in a water bath held at 80°C. The piston is slowly moved out to expand the volume of the cylinder to 20 L as the pressure inside the cylinder is monitored. A plot of the pressure versus volume for the system is shown in the figure above. Which of the following statements most closely indicates, with justification, the region of the curve where the equilibrium represented below occurs?



- (A) Region A, because the initial pressure inside the cylinder is equal to the pressure outside the cylinder, so there is no net force on the piston.
- (B) Region B, because the pressure inside the cylinder is equal to the vapor pressure of water at 80°C when both liquid and gas phases are present. ✓
- (C) Region C, because the water vapor is behaving according to the ideal gas law as expansion occurs.
- (D) Region D, because the pressure inside the cylinder has leveled off.

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4.



$$\Delta H^\circ = 162 \text{ kJ/mol}_{rxn}$$

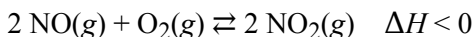
A sealed rigid vessel contains $\text{BaO}_2(s)$ in equilibrium with $\text{BaO}(s)$ and $\text{O}_2(g)$ as represented by the equation above. Which of the following changes will increase the amount of $\text{BaO}_2(s)$ in the vessel?

- (A) Removing a small amount of $\text{O}_2(g)$
- (B) Removing a small amount of $\text{BaO}(s)$
- (C) Adding He gas to the vessel

(D) Lowering the temperature



5.



Which of the following changes alone would cause a decrease in the value of K_{eq} for the reaction represented above?

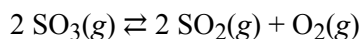
- (A) Decreasing the temperature

(B) Increasing the temperature



- (C) Decreasing the volume of the reaction vessel
- (D) Increasing the volume of the reaction vessel
- (E) Adding a catalyst

6.



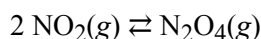
After the equilibrium represented above is established, some pure $\text{O}_2(g)$ is injected into the reaction vessel at constant temperature. After equilibrium is reestablished, which of the following has a lower value compared to its value at the original equilibrium?

- (A) K_{eq} for the reaction
- (B) The total pressure in the reaction vessel
- (C) The amount of $\text{SO}_3(g)$ in the reaction vessel
- (D) The amount of $\text{O}_2(g)$ in the reaction vessel

(E) The amount of $\text{SO}_2(g)$ in the reaction vessel



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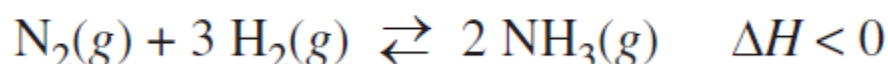


dark brown colorless

The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

7. A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is at equilibrium in a rigid reaction vessel. If the temperature of the mixture is decreased, then
- (A) $[\text{NO}_2]$ will increase and the mixture will turn a darker brown
 - (B) $[\text{N}_2\text{O}_4]$ will increase and the mixture will turn a lighter brown ✓
 - (C) $[\text{N}_2\text{O}_4]$ will decrease and the mixture will turn a lighter brown
 - (D) no change will be observed
-

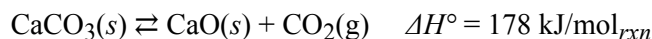
8.



$\text{NH}_3(g)$ was synthesized at 200°C in the presence of a powdered $\text{Os}(s)$ catalyst, leading to the equilibrium system represented above. Which of the following changes would result in more $\text{NH}_3(g)$ in the mixture after equilibrium is reestablished?

- (A) Replacing the powdered $\text{Os}(s)$ with a solid cube of $\text{Os}(s)$ of the same total mass
- (B) Increasing the temperature of the system to 250°C at constant pressure
- (C) Removing some $\text{H}_2(g)$
- (D) Adding some $\text{N}_2(g)$ ✓

9.



The reaction system represented above is at equilibrium. Which of the following will decrease the amount of $\text{CaO}(s)$ in the system?

- (A) Increasing the volume of the reaction vessel at constant temperature
 - (B) Lowering the temperature of the system ✓
 - (C) Removing some $\text{CO}_2(g)$ at constant temperature
 - (D) Removing some $\text{CaCO}_3(s)$ at constant temperature
-

Chapter 10 Part 2: Le Chatelier's Principle MCQ

10.



The synthesis of $\text{CH}_3\text{OH}(g)$ from $\text{CO}(g)$ and $\text{H}_2(g)$ is represented by the equation above. The value of K_c for the reaction at 483 K is 14.5.

Which of the following explains the effect on the equilibrium constant, K_c , when the temperature of the reaction system is increased to 650 K?

- (A) K_c will increase because the activation energy of the forward reaction increases more than that of the reverse reaction.
- (B) K_c will increase because there are more reactant molecules than product molecules.
- (C) K_c will decrease because the reaction is exothermic. ✓
- (D) K_c is constant and will not change.

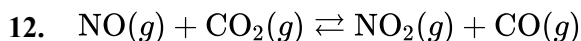
11.



The synthesis of $\text{CH}_3\text{OH}(g)$ from $\text{CO}(g)$ and $\text{H}_2(g)$ is represented by the equation above. The value of K_c for the reaction at 483 K is 14.5.

A mixture of $\text{CO}(g)$ and $\text{H}_2(g)$ is pumped into a previously evacuated 2.0 L reaction vessel. The total pressure of the reaction system is 1.2 atm at equilibrium. What will be the total equilibrium pressure of the system if the volume of the reaction vessel is reduced to 1.0 L at constant temperature?

- (A) Less than 1.2 atm
- (B) Greater than 1.2 atm but less than 2.4 atm ✓
- (C) 2.4 atm
- (D) Greater than 2.4 atm



Which of the following changes will increase the amount of $\text{NO}_2(g)$ in the equilibrium system represented by the equation above?

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- (A) Removing $\text{CO}(g)$ from the system ✓
- (B) Removing $\text{CO}_2(g)$ from the system
- (C) Increasing the volume of the reaction vessel
- (D) Decreasing the volume of the reaction vessel

13. $\text{X}_2(g) + \text{Y}_2(g) \rightleftharpoons 2 \text{XY}(g) \quad K_c = 3.0$

A mixture of $\text{X}_2(g)$, $\text{Y}_2(g)$, and $\text{XY}(g)$ is placed in a previously evacuated, rigid container and allowed to reach equilibrium at a constant temperature, as shown above. Which of the following sets of initial concentrations would lead to the formation of more product as the system moves toward equilibrium?

(A)	$[\text{X}_2]_{\text{initial}}$	$[\text{Y}_2]_{\text{initial}}$	$[\text{XY}]_{\text{initial}}$	✓
	0.40 M	0.40 M	0.20 M	

(B)	$[\text{X}_2]_{\text{initial}}$	$[\text{Y}_2]_{\text{initial}}$	$[\text{XY}]_{\text{initial}}$
	0.30 M	0.30 M	0.90 M

(C)	$[\text{X}_2]_{\text{initial}}$	$[\text{Y}_2]_{\text{initial}}$	$[\text{XY}]_{\text{initial}}$
	0.15 M	0.15 M	0.30 M

(D)	$[\text{X}_2]_{\text{initial}}$	$[\text{Y}_2]_{\text{initial}}$	$[\text{XY}]_{\text{initial}}$
	0.10 M	0.10 M	0.20 M

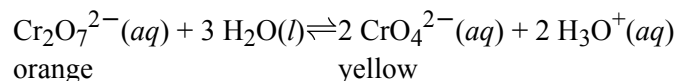
14. $\text{Fe}^{3+}(aq) + \text{SCN}^{-}(aq) \rightleftharpoons \text{FeSCN}^{2+}(aq)$
 Colorless Colorless Red

When colorless solutions containing $\text{Fe}^{3+}(aq)$ ions and $\text{SCN}^{-}(aq)$ ions are combined, a deep-red complex ion, $\text{FeSCN}^{2+}(aq)$ quickly forms, as shown in the net ionic equation above. Which of the following explains the observation that adding a few additional crystals of $\text{KSCN}(s)$ results in the red color of the solution becoming deeper?

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- (A) The added $\text{KSCN}(s)$ dissolves, disturbing the charge balance in the solution, causing $\text{Fe}(\text{SCN})_3$ to precipitate as a red solid.
- (B) The added $\text{KSCN}(s)$ dissolves, causing the solution to become saturated in SCN^- ions, which appear red at high concentrations.
- (C) The added $\text{KSCN}(s)$ dissolves, causing the reaction system to respond by producing more product to partially consume $\text{SCN}^-(aq)$ and reduce its concentration. ✓
- (D) The added $\text{KSCN}(s)$ dissolves, causing the reaction system to respond by forming more Fe^{3+} ions, which have a deep orange color at high concentrations.

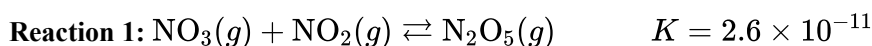
15.



The equilibrium system represented by the equation above initially contains equal concentrations of $\text{Cr}_2\text{O}_7^{2-}(aq)$ and $\text{CrO}_4^{2-}(aq)$. Which of the following statements correctly predicts the result of adding a sample of 6.0 M $\text{NaOH}(aq)$ to the system, and provides an explanation?

- (A) The mixture will become more orange because $\text{OH}^-(aq)$ will oxidize the Cr in $\text{CrO}_4^{2-}(aq)$.
- (B) The mixture will become more yellow because $\text{OH}^-(aq)$ will reduce the Cr in $\text{Cr}_2\text{O}_7^{2-}(aq)$.
- (C) The mixture will become more yellow because $\text{OH}^-(aq)$ will shift the equilibrium toward products. ✓
- (D) The color of the mixture will not change because $\text{OH}^-(aq)$ does not appear in the equilibrium expression.

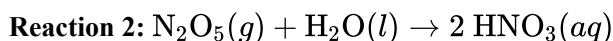
At night, an equilibrium reaction between two different nitrogen compounds generates N_2O_5 in the atmosphere, as represented below.



16. During the day, solar radiation is absorbed by $\text{NO}_3(g)$, resulting in its decomposition. Which of the following best explains whether the equilibrium concentration of $\text{N}_2\text{O}_5(g)$ in the atmosphere in the daytime is different from that in the nighttime, and why?
- (A) $[\text{N}_2\text{O}_5]$ will be higher during the day, because the decomposition of $\text{NO}_3(g)$ results in an increase in the rate of production of $\text{N}_2\text{O}_5(g)$.
- (B) $[\text{N}_2\text{O}_5]$ will be higher during the day, because $\text{NO}_2(g)$ will be in excess, leading to an increase in the rate of production of $\text{N}_2\text{O}_5(g)$.
- (C) $[\text{N}_2\text{O}_5]$ will be higher at night, because the decomposition of $\text{NO}_3(g)$ in the daytime will result in an increase in the rate of consumption of $\text{N}_2\text{O}_5(g)$ to reform $\text{NO}_3(g)$. ✓
- (D) $[\text{N}_2\text{O}_5]$ will be about the same at nighttime and daytime, because the amount of $\text{NO}_2(g)$ will not be changed and the equilibrium will not be affected.

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17. In the atmosphere, small water droplets are suspended in the air, forming an aerosol. $\text{N}_2\text{O}_5(g)$ can form $\text{HNO}_3(aq)$ under these conditions, as shown in reaction 2, represented below.



Which of the following predicts the effect that the formation of $\text{HNO}_3(aq)$ will have on the equilibrium shown in reaction 1, and why?

- (A) The equilibrium of reaction 1 will shift toward the formation of more product, because $\text{N}_2\text{O}_5(g)$ is removed when it reacts to form $\text{HNO}_3(aq)$. ✓
- (B) The equilibrium of reaction 1 will shift toward the formation of more product, because $\text{H}_2\text{O}(l)$ acts as a catalyst for reaction 1.
- (C) The equilibrium of reaction 1 will shift toward the formation of more reactants, because the product $\text{N}_2\text{O}_5(g)$ is removed when it reacts to form $\text{HNO}_3(aq)$.
- (D) The equilibrium of reaction 1 is not affected, because neither $\text{H}_2\text{O}(l)$ nor $\text{HNO}_3(aq)$ are gases.

18. $\text{Cl}_2(aq) + 2 \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq) + \text{HOCl}(aq)$

$$K_c = \frac{[\text{H}_3\text{O}^+][\text{Cl}^-][\text{HOCl}]}{[\text{Cl}_2]} = 4.8 \times 10^{-4}$$

The equilibrium reaction between $\text{Cl}_2(aq)$ and $\text{H}_2\text{O}(l)$ at 25°C is represented by the chemical equation shown above. If a solution at equilibrium at 25°C is diluted with distilled water to twice its original volume, which of the following gives the value for Q_c and predicts the response by the system immediately after dilution?

- (A) $Q_c = 4K_c$, and the rate of the reverse reaction will be greater than the rate of the forward reaction.
- (B) $Q_c = 4K_c$, and the rate of the forward reaction will be greater than the rate of the reverse reaction.
- (C) $Q_c = \frac{K_c}{4}$, and the rate of the reverse reaction will be greater than the rate of the forward reaction.
- (D) $Q_c = \frac{K_c}{4}$, and the rate of the forward reaction will be greater than the rate of the reverse reaction. ✓

19. $\text{N}_2(g) + 3 \text{H}_2(g) \rightleftharpoons 2 \text{NH}_3(g)$

A 2.0 L reaction vessel contains an equilibrium mixture of the system represented above. The partial pressures of the components of the mixture at equilibrium are listed in the table below.

P_{N_2}	P_{H_2}	P_{NH_3}
0.30 atm	0.50 atm	0.20 atm

Which of the following is the best estimate of the total pressure of the system when the volume of the container is reduced to 1.0 L at a constant temperature?

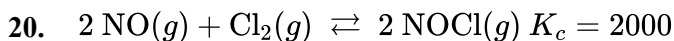
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(A) $P_{total} = 1.00 \text{ atm}$

(B) $1.00 \text{ atm} < P_{total} < 2.00 \text{ atm}$

(C) $P_{total} = 2.00 \text{ atm}$

(D) $P_{total} > 2.00 \text{ atm}$



A mixture of $\text{NO}(g)$ and $\text{Cl}_2(g)$ is placed in a previously evacuated container and allowed to reach equilibrium according to the chemical equation shown above. When the system reaches equilibrium, the reactants and products have the concentrations listed in the following table.

Species	Concentration (M)
$\text{NO}(g)$	0.050
$\text{Cl}_2(g)$	0.050
$\text{NOCl}(g)$	0.50

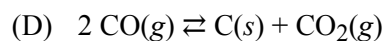
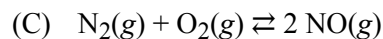
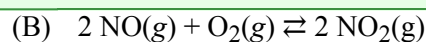
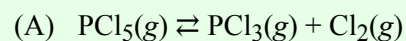
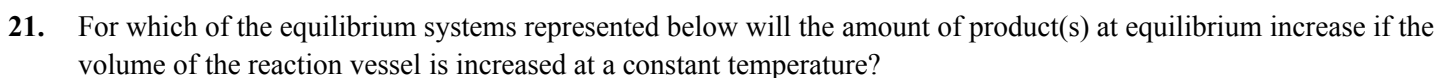
Which of the following is true if the volume of the container is decreased by one half?

(A) $Q = 100$, and the reaction will proceed toward reactants.

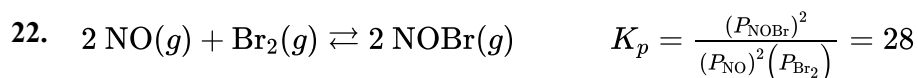
(B) $Q = 100$, and the reaction will proceed toward products.

(C) $Q = 1000$, and the reaction will proceed toward reactants.

(D) $Q = 1000$, and the reaction will proceed toward products.



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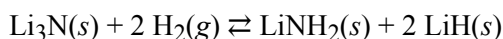
Substance	Initial Partial Pressure (torr)
NO	10.
Br ₂	2.0
NOBr	80.

The system represented by the equation above is allowed to establish equilibrium. The initial pressures of the substances are given in the table. Which of the following explains what the system will do as it approaches equilibrium?

- (A) $Q = \frac{(80.)^2}{(10.)^2(2.0)} > K_p$ and equilibrium will be approached by producing NOBr because the forward reaction is faster than the reverse reaction.
- (B) $Q = \frac{(80.)}{(10.)^2(2.0)} < K_p$ and equilibrium will be approached by producing NOBr because the forward reaction is faster than the reverse reaction.
- (C) $Q = \frac{(80.)^2}{(10.)^2(2.0)} < K_p$ and equilibrium will be approached by consuming NOBr because the reverse reaction is faster than the forward reaction.
- (D) $Q = \frac{(80.)^2}{(10.)^2(2.0)} > K_p$ and equilibrium will be approached by consuming NOBr because the reverse reaction is faster than the forward reaction. ✓

23. For which of the following reaction mixtures at equilibrium would reducing the volume of the container at constant temperature cause the equilibrium to shift toward the products?

- (A) $\text{SrCO}_3(s) \rightleftharpoons \text{SrO}(s) + \text{CO}_2(g)$
- (B) $\text{C}(s) + \text{CO}_2(g) \rightleftharpoons 2 \text{CO}(g)$
- (C) $\text{CO}(g) + \text{H}_2\text{O}(g) \rightleftharpoons \text{CO}_2(g) + \text{H}_2(g)$
- (D) $\text{PCl}_5(g) \rightleftharpoons \text{PCl}_3(g) + \text{Cl}_2(g)$
- (E) $2 \text{NO}_2(g) \rightleftharpoons \text{N}_2\text{O}_4(g)$ ✓



$\Delta H^\circ = -192 \text{ kJ/mol}_{rxn}$

Because pure H₂ is a hazardous substance, safer and more cost effective techniques to store it as a solid for shipping purposes have been developed. One such method is the reaction represented above, which occurs at 200°C.

24. The amount of H₂(g) present in a reaction mixture at equilibrium can be maximized by

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- (A) increasing the temperature and increasing the pressure by decreasing the volume
- (B) increasing the temperature and decreasing the pressure by increasing the volume ✓
- (C) decreasing the temperature and increasing the pressure by decreasing the volume
- (D) decreasing the temperature and decreasing the pressure by increasing the volume
-

25.

Which of the following changes to the equilibrium system represented above will increase the quantity of HI(g) in the equilibrium mixture?

- I. Adding $\text{H}_2(\text{g})$
II. Increasing the temperature
III. Decreasing the pressure

- (A) I only
(B) III only

(C) I and II only ✓

- (D) II and III only
(E) I, II, and III

26. $\text{HgO}(\text{s}) + \text{H}_2\text{O} \rightleftharpoons \text{HgI}_4^{2-} + 2\text{OH}^-$

Consider the equilibrium above. Which of the following changes will increase the concentration of HgI_4^{2-} ?

(A) Increasing the concentration of OH^-

(B) Adding 6 M HNO_3 ✓

- (C) Increasing the mass of HgO present
(D) Increasing the temperature
(E) Adding a catalyst

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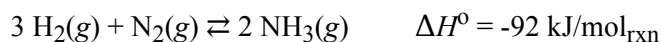
At a certain time, the partial pressure of the gases in the reaction mixture represented above had the values shown in the table below.

P_{CO}	P_{H_2}	P_{CH_4}	$P_{\text{H}_2\text{O}}$
0.02 atm	1.0 atm	0.4 atm	0.4 atm

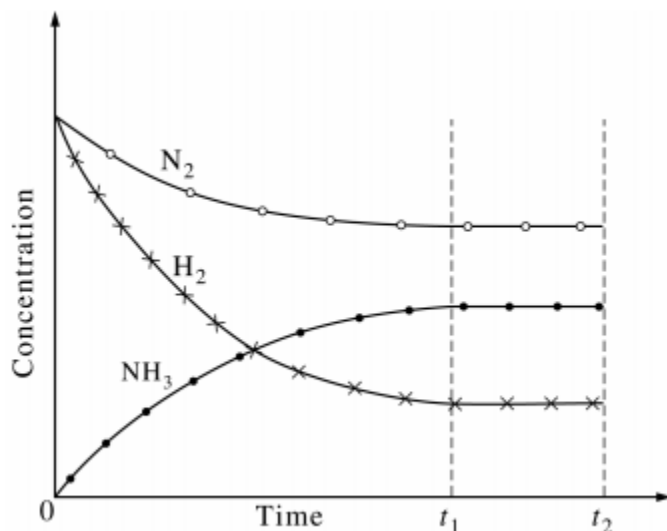
Based on the information above, which of the following occurred as the reaction mixture moved toward equilibrium?

- (A) More $\text{CH}_4(g)$ was produced because the rate of the forward reaction was higher than the rate of the reverse reaction. ✓
- (B) More $\text{H}_2(g)$ was produced because the rate of the reverse reaction was higher than the rate of the forward reaction.
- (C) More $\text{CH}_4(g)$ was produced because the total pressure of the $\text{H}_2(g)$ and $\text{CO}(g)$ combined was higher than that of the products; thus the reaction shifted to the side with the fewest number of moles of gas.
- (D) More $\text{H}_2(g)$ was produced because the pressure of $\text{CO}(g)$ was the least and the reaction shifted to the side with the smaller number of moles of gas.

H_2 gas and N_2 gas were placed in a rigid vessel and allowed to reach equilibrium in the presence of a catalyst according to the following equation.

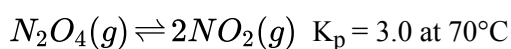


The diagram below shows how the concentrations of H_2 , N_2 , and NH_3 in this system changed over time.



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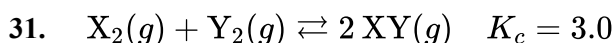
28. More NH_3 gas is added to the system at time t_2 while the temperature is held constant. Which of the following will most likely occur?
- (A) The value of the equilibrium constant will increase
 - (B) The value of the equilibrium constant will decrease.
 - (C) The total pressure in the container will decrease.
 - (D) The amount of N_2 will increase. ✓
 - (E) The amount of H_2 will decrease.
-



colorless brown

A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C , as represented above.

29. Which of the following best predicts how the partial pressures of the reacting species will be affected if a small amount of $\text{Ar}(g)$ is added to the equilibrium mixture at constant volume?
- (A) P_{NO_2} will decrease and $P_{\text{N}_2\text{O}_4}$ will increase.
 - (B) P_{NO_2} will increase and $P_{\text{N}_2\text{O}_4}$ will decrease.
 - (C) Both P_{NO_2} and $P_{\text{N}_2\text{O}_4}$ will decrease.
 - (D) No change will take place. ✓
30. Which of the following statements best helps to explain why the contents of the tube containing the equilibrium mixture turned a lighter color when the tube was placed into an ice bath?
- (A) The forward reaction is exothermic.
 - (B) The forward reaction is endothermic. ✓
 - (C) The ice bath lowered the activation energy.
 - (D) The ice bath raised the activation energy.
-



A mixture of $\text{X}_2(g)$, $\text{Y}_2(g)$, and $\text{XY}(g)$ is placed in a previously evacuated, rigid container and allowed to reach equilibrium at a constant temperature, as shown above. Which of the following sets of initial concentrations would lead to the formation of more product as the system moves toward equilibrium?

Chapter 10 Part 2: Le Chatelier's Principle MCQ

(A)	$[X_2]_{initial}$	$[Y_2]_{initial}$	$[XY]_{initial}$	
	0.50 M	0.45 M	0.20 M	

(B)	$[X_2]_{initial}$	$[Y_2]_{initial}$	$[XY]_{initial}$
	0.10 M	0.10 M	0.90 M

(C)	$[X_2]_{initial}$	$[Y_2]_{initial}$	$[XY]_{initial}$
	0.30 M	0.10 M	0.30 M

(D)	$[X_2]_{initial}$	$[Y_2]_{initial}$	$[XY]_{initial}$
	0.10 M	0.80 M	0.75 M

32. Reaction A: $4 \text{HCl}(g) + \text{O}_2(g) \rightleftharpoons 2 \text{Cl}_2(g) + 2 \text{H}_2\text{O}(g)$

Reaction B: $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$

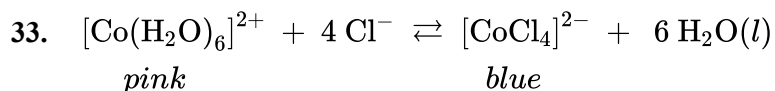
Reaction C: $\text{H}_2(g) + \text{I}_2(g) \rightleftharpoons 2 \text{HI}(g)$

Reaction D: $2 \text{NH}_3(g) \rightleftharpoons \text{N}_2(g) + 3 \text{H}_2(g)$

The reactions represented above are carried out in sealed, rigid containers and allowed to reach equilibrium. If the volume of each container is reduced from 1.0 L to 0.5 L at constant temperature, for which of the reactions will the amount of product(s) be increased?

(A) Reaction A	
(B) Reaction B	
(C) Reaction C	
(D) Reaction D	

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The solvent in the nonaqueous system represented above is 2-propanol. If a small amount of water is added to the system, how will the color of the solution be affected, and why?

(A) The solution will become more pink because more $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ will be produced. ✓

(B) The solution will become more pink because the pH of the solution will increase.

(C) The solution will become more blue because more $[\text{CoCl}_4]^{2-}$ will be produced.

(D) The solution will become more blue because the pH of the solution will decrease.

34.

Compound	$\Delta H_{\text{solution}}$ (kJ / mol)
$\text{NaF}(s)$	+0.91
$\text{MgF}_2(s)$	−18

Based on the information in the table above, the solubility of which compound, if any, will increase if the water temperature is increased?

(A) $\text{NaF}(s)$ only ✓

(B) $\text{MgF}_2(s)$ only

(C) Both $\text{NaF}(s)$ and $\text{MgF}_2(s)$

(D) The solubility of neither $\text{NaF}(s)$ nor $\text{MgF}_2(s)$ will increase.